120 Days of Quantitative Finance

Month 5: Advanced Quant Systems

Week 17

Day 119: End-to-End Quant Trading Pipeline

1 Introduction

A complete quantitative trading system integrates signal generation, volatility modeling, and risk management.

2 Pipeline Overview

1. Data Collection
2. Feature Engineering
3. Signal Generation
4. Volatility Estimation
5. Position Sizing
6. Risk Management
7. Execution

3 Data and Returns

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

4 Signal Generation

Predict returns:

$$\hat{r}_{t+1} = f(X_t)$$

Signal:

$$\text{Signal}_t = \begin{cases} +1 & \text{if } \hat{r}_{t+1} > \theta \\ -1 & \text{if } \hat{r}_{t+1} < -\theta \\ 0 & \text{otherwise} \end{cases}$$

5 Volatility Estimation

Using GARCH:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

6 Position Sizing

Risk-adjusted position:

$$w_t = \frac{\text{Signal}_t}{\sigma_t}$$

7 Portfolio Return

$$R_t = w_t \cdot r_t$$

8 Risk Management

- Stop-loss constraints
- Maximum drawdown limits
- Volatility targeting

9 Sharpe Ratio

$$\text{Sharpe} = \frac{\mathbb{E}[R_t]}{\sigma(R_t)}$$

10 Key Insight

- Signal determines direction
- Volatility determines size
- Risk controls survival

11 Conclusion

A robust trading system requires integrating prediction, volatility modeling, and disciplined risk management.